Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Getting Started

- Course Introduction
  Video • 1 min
- **Course Overview**
  Reading • 2 min
- Syllabus
  Reading • 10 min
- Meet Your Instructor
  Video • 1 min
- Meet Your Fellow Learners
  Discussion Prompt • 10 min
- Academic Integrity
  Reading • 1 min

Lesson 1: Introduction to Linear Algebra

- Introduction to Machine Learning
  Reading • 25 min
- Introduction to Linear Algebra
  Reading • 4 min
- Why Linear Algebra and Mathematics?
  Reading • 2 min
- Notation
  Reading • 5 min

Lesson 2: Systems of Linear Equations

- Foundational Concepts of Systems of Linear Equations
  Reading • 20 min
- [H5P] System of Linear Equations
  Ungraded App Item • 20 min
- Solved Example and Recommended Resources
  Reading • 20 min
- Check Your Knowledge: System of Linear Equations
  Graded Assignment • 16 min

Lesson 3: Matrices

- Matrices and Matrix Operations
  Reading • 20 min
- Matrices
  Video • 8 min
- Helpful Resources and Solved Examples
  Reading • 10 min
- Check Your Knowledge: Matrices
  Graded Assignment • 16 min

Lesson 4: Vector Spaces

- Foundations of Vector Spaces: Operations and Subspaces
  Reading • 20 min
- Vector Space
  Video • 7 min
- Vector Space Properties
  Reading • 10 min
- Subspace
  Reading • 1 min
- Check Your Knowledge: Vector Spaces
  Graded Assignment • 14 min

Lesson 5: Orthogonal Complement and Projection

- Introduction to Orthogonal Complement
  Reading • 20 min
- Introduction to Orthogonal Projections
  Reading • 40 min

Course Overview

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This course is a segment of the broader foundations course, focusing on key concepts in linear algebra, probability, and optimization techniques. You'll start with an introduction to linear algebra, covering systems of linear equations and matrix operations. The course then delves into vector spaces, linear independence, and linear mappings. You'll explore orthogonal complements and projections, eigenvalues and eigenvectors, and advanced topics like Cholesky decomposition and singular value decomposition. Additionally, the course includes modules on probability, such as the sum rule, product rule, Bayes' theorem, and Gaussian distribution. You'll also learn about inference techniques, model selection methods including cross-validation, and an introduction to optimization techniques.

It is highly recommended that you take the pre-calculus test available on the MDTP website. This test is designed to assess your foundational knowledge and ensure you're well-prepared for the advanced topics covered in this course. The test typically takes about 60 to 90 minutes to complete and will provide valuable insights into areas where you may need to focus your studies. By taking this test, you'll be better equipped to tackle the challenging concepts in linear algebra, probability, and optimization techniques that we will explore together.

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